LINKEDIN JOB

JOB MARKET INTELLIGANCE

Group # 1

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Problem & Motivation

Problem Statement

How can distributed big data processing be leveraged to extract meaningful insights — including job demand patterns, required skills, and salary predictions — from large-scale LinkedIn job postings?

Context

Millions of job postings generated daily — hard to analyze at scale

Stakeholders

Job Seekers · Recruiters
HR Teams · Researchers
Career Counselors

Dataset Overview

Source: LinkedIn Job Listings — Hugging Face
URL: huggingface.co/datasets/Moheez2611/linkedin_job_listings
Domain: Human Resources & Labor Market Analytics

~123K	34	3 – 4 / 2024
Row	Column	Time span

Attribute	Description
normalized_salary	Target variable – standardized annual salary
formatted_experience_level	Seniority level (Entry, Mid-Senior, Director...)
formatted_work_type	Work arrangement (Full-time, Contract...)
state / location	Geographic location extracted from posting
remote_allowed	Whether remote work is permitted

Data Quality & Prepsocessing

MAIN DATA QUALITY ISSUES

- **High missingness in:**
  - closed_time (99.13%)
  - skills_desc (98.03%)
  - med_salary (94.93%)
- **Salary inconsistencies and extreme outliers**
- **High-cardinality categorical fields**
  - (title, location)

KEY PREPROCESSING STEPS

- **Dropped highly sparse columns**
- **Imputed missing values:**
  - categorical → "Unknown"
  - numeric → 0.0
  - salary → median values
- **Reduced dimensionality by:**
  - extracting state
  - removing redundant features
- **Applied:**
  - StringIndexer
  - OneHotEncoder
  - StandardScaler

Metric	Before	After
Rows	123,849	123,394
Columns	34	7
Sparse Columns	3	0
Salary Outliers	455	0

RDD EXPLORATION

Filter

Isolate remote jobs from the dataset to analyze flexible work opportunities

reduceByKey

Compute average salary per state for geographic salary analysis

sortByKey

Rank states by average salary to identify highest-paying regions

Interpretation

- Remote jobs represented only 12.3% of total postings
- Geographic location strongly influenced salary levels
- Metropolitan regions consistently showed the highest compensation
- Results helped identify important salary prediction features such as location and work type

Scala Snippet

```
val stateTotalsRDD =
  stateSalaryRDD.reduceByKey((a, b) =>
    (a._1 + b._1, a._2 + b._2)
)

val sortedRDD = stateAvgRDD
    .map { case (state, avg) => (avg, state) }
  .sortByKey(false)
```

Result Summary

State	Avg Salary
San Francisco Bay Area	$119,270
New York City Metro Area	$106,076
Greater Seattle Area	$105,693

SQL ANALYTICS HIGHLIGHTS

Average salary by experience level

Executive and Director roles had the highest average salaries

Top states by average salary

San Francisco Bay Area and New York City ranked highest

Salary ranking within experience levels

Revealed extreme salary outliers and large salary variation within the same seniority group

Result Summary

Experience Level	Avg Salary
Executive	$119,888
Director	$113,012
Mid-Senior Level	$92,309

Interpretation

- Experience level and geographic location were major salary predictors
- SQL analytics revealed salary outliers exceeding regional averages
- Findings directly supported the ML task of salary prediction by identifying the most influential features

ML Approach

PROBLEM TYPE

Supervised Regression
(Predicting Continuous Salary)

MODEL CHOSEN

Random Forest (Ensemble) &
Gradient Boosted Trees (GBT)

KEY FEATURE

Job Title, State, Work Type,
Experience Level, Remote
Allowed

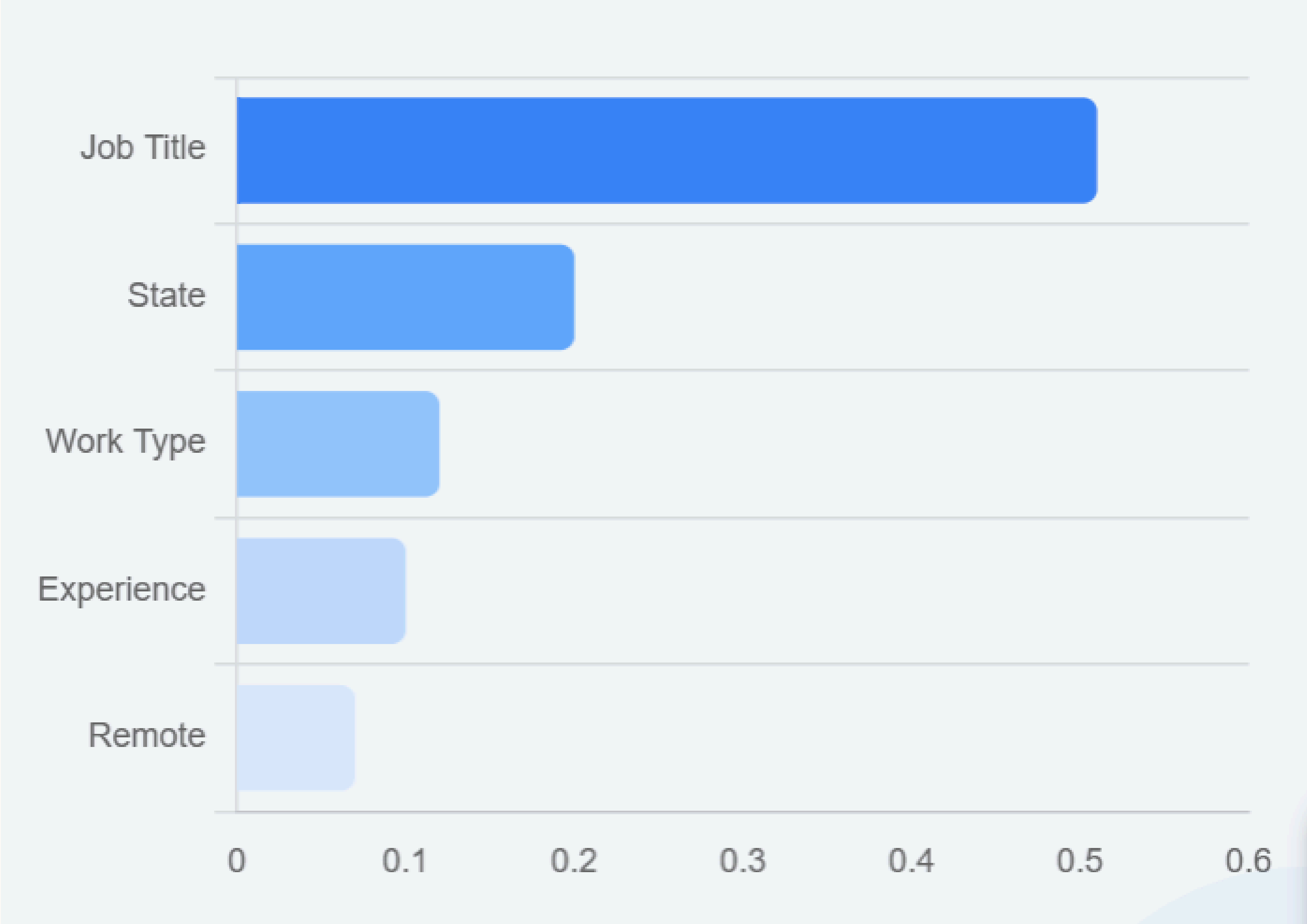
TRAIN/TEST SPLIT

80% Training / 20% Testing |
Seed: 42

Results & Model Performance

Model	RMSE ($)	MAE ($)	R^2 Score
Baseline (Mean)	62,016	44,405	-0.0005
Random Forest	53,987	35,999	0.2418
GBT (Winner)	**52,944**	**34,507**	**0.2708**

Interpretation: GBT outperformed the baseline by 14.6%. R^2 of 0.27 is a solid indicator for real-world salary data.



INTERPRETATION & INSIGHTS

- **Top 3 important features:** Job Title, followed by State and Work Type.
- **Where the model performs well:** Estimating salaries for common job titles.
- **Where it struggles**: Predicting salary for rare roles with low data frequency.
- **Practical implications:** Recruiters can benchmark competitive salary ranges; job seekers can identify location-based pay disparities.

LIMITATIONS & FUTURE WORK

Limitations

- Missing skills data (98% nulls)
- Limited temporal window (2 months)
- No company revenue data

Future Work

- Spark NLP for skill extraction
- Real-time streaming integration
- Hyperparameter Grid Search

CONCLUSION

Main Achievements

- Built end-to-end Spark pipeline for 123k+ records.
- GBT model improved salary prediction accuracy by 14.6%.

Key Takeaway

Data transparency enables informed pay decisions for job seekers and recruiters alike.

THANK *You*

Any Questions?